

Deterministic differentiable exact-cone contact on GPUs: bit-identical adjoints, batched routing and system identification

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21 September 2026, revision 2

Abstract

Differentiating a contact step requires agreement between the forward contact law and the equations used for its sensitivity. Parallel accumulation introduces an additional concern: changes in floating point summation order can change the contact regime on which the derivative is evaluated. We combine proximal ADMM, a Coulomb cone projection and a de Saxcé correction with fixed-regime implicit differentiation. Integer accumulation makes the scatter independent of thread order, conditional on representable sums. The sensitivity calculation reuses a symmetric contact operator and solves a separate affine system for the non-associated friction correction.

Robot stance tests agree with analytical constrained dynamics, and independent implicit differentiation checks slipping configurations. A deterministic indexing defect demonstrates why repeated bytes alone cannot validate a derivative. Batch measurements separate execution rate from the fraction of environments reaching the contact tolerance. In simulated object identification, observation noise limits parameter recovery, while stronger excitation improves mass estimation without meeting the friction target or respecting the robot’s torque limits. The results establish repeatable contact sensitivities in the tested regimes; they do not establish a verified reverse-mode implementation, convergence for dense piles, or identification on hardware.

1 Introduction

A contact derivative used in estimation must describe how the solved contact law changes with the unknown parameter. Replacing Coulomb friction by a convex relaxation changes this map even when both numerical solves converge. Conversely, retaining the Coulomb law does not make a finite-budget iterate or its derivative accurate. The distinction becomes consequential near separation, where small numerical changes can alter the contact regime.

Complementarity formulations connect rigid contact to optimization without prescribing a contact sequence in advance, as demonstrated by Posa, Cantu and Tedrake [1]. Analytical differentiation of rigid body dynamics supplies another part of this computation [2], and Crocoddyl provides an optimal-control framework using those derivatives [5]. Proximal methods address compliant and rigid contact within a common formulation [3]. Our contribution concerns their numerical implementation: a contact operator with deterministic accumulation, its fixed-regime sensitivity, and experiments that assess derivative accuracy independently of execution repeatability. We make no priority claim for the individual solver ingredients.

A sliding box distinguishes the contact laws through their normal separation. Robot stance tests then check the sensitivity where unilateral and bilateral models coincide, while slipping tests exercise the non-associated correction. These comparisons separate formulation error from derivative error. Object identification asks a further question: whether contact observations constrain the unknown parameters. That experiment uses a separate implicit derivative implementation, so its estimation errors do not validate an integrated GPU sensitivity pipeline.

2 Contact equations and stopping conditions

Let J map generalized velocities to contact velocities, M be the generalized mass matrix, and λ collect contact impulses in normal-tangential order. Define the Delassus operator G and the free contact velocity b by

$$[G = JM^{-1}J, b = Jv_{-}, v^{+} = v_{-} + M^{-1}J^{\top} \lambda.]$$

A speculative offset, when present, is included in b . The regularized contact velocity is $u = H\lambda + b$ with $H = G + E$, where E contains isotropic compliance blocks. Compliance has inverse-mass units. Positive compliance changes the contact problem and can select an impulse in a hyperstatic configuration; it is not merely a preconditioner. With compliance, u includes $E\lambda$ and must be distinguished from the physical contact velocity Jv^{+} . The rigid equations correspond to vanishing E .

For positive friction coefficient μ , define

$$[K_{-} = \{(\lambda_{-}, t) : \lambda_{-} \geq 0, |t| \leq \mu \lambda_{-}\}, c(u) = (c|u|, 0, 0).]$$

The de Saxcé formulation [4] is

$$[K_{-}c(u) - c(u)K_{-}^{*}, K_{-}^{*} = K_{-}^{-1}.]$$

The frictionless limit uses $\lambda_{-} = 0$ with scalar normal complementarity, rather than the reciprocal-friction cone. “Exact cone” refers to this friction formulation and its projection, not to an exact solution after a finite iteration budget.

Lemma (closed sliding contact). Suppose the normal impulse is positive and the tangential velocity is nonzero. Write $d = u_{-} / \|u_{-}\|$. Coulomb slip gives $\lambda_{-} = -\mu \lambda_{-} d$ and $u_{-} = 0$; removing Γ instead gives $u_{-} = \mu \|u_{-}\| d$.

Proof. Substitution into $\lambda^{\top}(u + \Gamma) = 0$ cancels tangential dissipation against the normal correction. Division by the positive normal impulse gives the Coulomb relation. Without Γ , the tangential contribution remains, giving the relaxed normal velocity. For rigid contact without a speculative offset, multiplying the normal velocity by the step size gives the local separation increment. With compliance or an offset, the regularized normal velocity is not itself the physical separation rate. This argument concerns a fixed contact branch, not a rocking trajectory with changing contacts.

For frozen Γ , ADMM solves the cone-constrained quadratic

$$[\frac{1}{2}z^{\top}Kz + (b + \Gamma)^{\top}z.]$$

With unscaled dual variable γ , the ADMM updates are

$$[\begin{aligned} (H + \rho I)x &= -(b + \Gamma + \gamma - \rho z), \quad z = \Pi_{K_{-}\mu}(x + \gamma/\rho), \\ \gamma &\leftarrow \gamma + \rho(x - z). \end{aligned}]$$

Refreshing Γ supplies the outer nonlinear correction. Convergence of the frozen quadratic alone does not establish convergence of the Coulomb problem. The large-scene path applies G through a body scatter and contact gather; the articulated batch path instead stores dense operators for its small contact systems. A block Jacobi preconditioner supports the inner conjugate-gradient solve. A fixed inner budget and a refresh schedule are part of the algorithm being measured.

We assess contact convergence with the natural map, rather than the ADMM splitting residual:

$$[\begin{aligned} \alpha_c &= \|H_{cc}\|^{-1}, \quad r_{\text{nat}}(\lambda) = \max_c \left| \lambda_c - \Pi_{K_{-}\mu_c}(\alpha_c[u_c + \Gamma_c(u)]) \right| \\ &\end{aligned}]$$

Here α is distinct from the ADMM penalty ρ . The residual is an absolute impulse discrepancy, expressed in N-s for the translational contact coordinates used here. It is not divided by impulse magnitude. The implementation substitutes a unit scale if a diagonal block has zero norm. For the relaxed model Γ is suppressed in both the solve and this diagnostic. A batch environment is frozen only when its checked residual is below the tolerance; exhausting the iteration budget is not a successful stop. Tables D and E distinguish the stopping threshold from the executed iteration count. Reference residuals in Table C describe the states supplied to the derivative test, not a separate guarantee that every forward batch reaches those states.

3 Fixed-regime sensitivity

The derivative is local to the contact regime. Open contacts impose zero impulse. Sticking contacts impose zero regularized contact velocity, and slipping contacts impose

$$[u_{n,c}=0, \{t,c\}+c\{n,c\}d_c=0, d_c=u\{t,c\}/|u\{t,c\}|.]$$

These equations use the same H as the forward solve. For a parameter perturbation, write $h=\delta H\lambda+\delta b$, so that $\delta u=H\delta\lambda+h$. Changes in mass, geometry and compliance belong in h whenever they depend on that parameter. The post-step velocity sensitivity also differentiates the free velocity and the impulse-to-velocity map; $\delta\lambda$ alone is insufficient for a position or mass derivative.

Proposition (implicit sensitivity). Collect the branch equations in $R(\lambda,\theta)$. If the regime persists, R is continuously differentiable in a neighborhood of the state, slip speeds are nonzero, and the impulse Jacobian is nonsingular, then

$$[R_{\lambda}=-R.]$$

For a differentiable scalar objective ℓ , its mathematical adjoint satisfies

$$[R_{\ell}^p = -pR_{\ell}.]$$

Proof. Differentiate $R(\lambda(\theta),\theta)=0$ and solve for the impulse variation. Substitute it into the chain rule for ℓ and eliminate the variation using the transposed system. The assumptions exclude regime transitions and singular impulse distributions. In particular, a pseudo-inverse at a singular configuration does not establish a unique physical impulse derivative.

The measured implementation computes parameter-direction sensitivities. Its reuse of symmetric linear algebra can be derived without claiming that the complete Coulomb Jacobian is symmetric. At a slipping contact with positive normal impulse and positive friction coefficient, let $d_{\perp}$ be a unit tangent orthogonal to d . Write $\delta\lambda=Zy+p_0$, where p_0 accounts for the movement of the cone boundary under $\delta\mu$. The local blocks are

$$[Z_s = \begin{bmatrix} 1 & 0 & 0 \\ \mu d & d_{\perp} & 0 \end{bmatrix}, D_s = \begin{bmatrix} 0 & 1 \\ 0 & -\mu d \end{bmatrix}, p_{0,s} = \begin{bmatrix} 0 \\ -\mu d \end{bmatrix}.]$$

Open contacts use a zero basis and identity D . Sticking contacts use an identity basis and zero D . The boundary-motion term vanishes in both cases. Let W select $d^T\delta u_t$ at slipping contacts, and let C inject $\mu\sigma$ into the first reduced coordinate of each such contact. Define

$$[A=Z^THZ+D, g=Hp_0+h.]$$

The coupled equations executed by the sensitivity method are

$$[Ay=-Z^Tg-C_{\ell}, \ell=W(HZy+g), \ell=Zy+p_0.]$$

To verify the reduction, differentiate the slip direction as $\delta d=(I-dd^T)\delta u_t/\|u_t\|$. The tangential component orthogonal to d gives the curvature term in D . The first reduced component is $\delta u_n-\mu d^T\delta u_t$; adding $\mu\sigma$ recovers $\delta u_n=0$. Thus the correction is needed even though A is symmetric.

Eliminating y gives an affine correction map:

$$\begin{aligned} \sigma &= a + B\sigma, \\ a &= Wg - WHZA^{-1}Z^{\top}g, \\ B &= -WHZA^{-1}C. \end{aligned}$$

The implementation evaluates the zero input and basis inputs, solves $(I - B)\sigma = a$, and recovers $\delta\lambda$. This derivation assumes nonsingular A and $I - B$ and exact inner solves; truncated conjugate gradients approximate the map. A poorly conditioned correction system can amplify inner-solve error. Applying A with conjugate gradients therefore neither solves the complete adjoint equation nor verifies a reverse-mode implementation. The reported direction rates measure the sensitivity computation just described.

Corollary (friction information in strict stick). If the other model quantities are independent of a contact’s friction coefficient, that coefficient has zero local sensitivity while the contact remains strictly sticking and the branch Jacobian is nonsingular.

Proof. The sticking equations contain no friction coefficient. Their parameter right-hand side therefore vanishes, and nonsingularity gives zero impulse variation. Cone feasibility still gives an inequality constraint on the coefficient; the corollary concerns a local derivative rather than the absence of all information.

4 Repeatability and derivative validation

Integer accumulation addresses the order of the contact scatter. For scale S , contributions are converted to integers, summed, and converted back before the contact gather. Other reductions use a prescribed reduction layout. The scale is recomputed as a power of two from the maximum component magnitude and a host-computed bound on the scatter coefficients. This reduces the conversion quantum as the Krylov residual shrinks. This remains a quantized operator; no finite-termination theorem for exact-arithmetic conjugate gradients follows from it.

Lemma (order independence). For fixed converted contributions, if every intermediate integer sum is representable, the scatter result is independent of thread order.

Proof. Integer addition agrees with addition in the integers under the stated range condition, so associativity and commutativity apply to every ordering. Conversion back from the same final integer yields the same represented result. A bound on the sum of absolute converted contributions suffices to check representability of every ordering. The lemma does not establish that bound for arbitrary scenes or accuracy of the converted contributions.

The tested compiler configuration disables floating point fusion and fast-math transformations. Host-built inputs and device outputs must nevertheless be distinguished: a change in host linear algebra can change the input bytes before the deterministic scatter is reached. Table F reports cross-device equality for the recorded batch outputs, including sensitivities. Those runs are separate from the corrected-kernel accuracy tests in Table C. Equal hashes therefore establish repetition of the recorded computation, without establishing derivative accuracy or a guarantee across compilers and architectures.

Table A tests a different property. The exact and relaxed sliding cases use the same geometry, time step and iteration budget, so the normal-gap difference isolates the friction formulation within that code. A cross-engine comparison would also change contact generation and time integration. The slip lemma explains the local separation mechanism, while the trajectory maximum remains a measured quantity.

Table B compares with Pinocchio in a bilateral sticking stance at rest. With stabilization disabled and compatible contact frames, impulse and acceleration formulations obey $\lambda = \Delta t f$ and $v^+ = \Delta t a$. The same scaling applies to torque derivatives. The position derivatives compose the contact

sensitivity with finite differences of the reference library’s kinematic and inertial quantities; no finite difference passes through the contact solve. Away from a compatible initial velocity, imposing zero contact acceleration and imposing zero post-step contact velocity are different maps. We therefore do not treat their velocity derivatives as interchangeable. Slipping contacts require the independent Coulomb reference rather than a bilateral force that may violate the cone.

The relative error in Table B is the maximum absolute entrywise difference divided by the maximum absolute reference entry. Table C uses the maximum of the reference and computed entry magnitudes in the denominator, with zero reported for equal zero arrays. The latter test spans every impulse right-hand-side and friction-coefficient direction, comparing the batch implementation against a separately implemented CPU differentiation of the branch equations at the same regularized state. The reference states and contact equations are shared; this is an implementation check, not validation against an independent physical model.

The faulty sensitivity kernel reproduced its own bytes while returning incorrect slipping derivatives. The faulty kernel evaluated the nominal zero input of the affine correction map with a nonzero correction at a slipping contact. The resulting error contaminated the reconstructed map even though repeated execution returned the same bytes. Correctly zeroing that evaluation restores the agreement shown in Table C. Partial-slip cases expose the mechanism: moving the sticking contact from the first position to the last changes whether the erroneous write affects a slipping contact. Neither the contact residual nor repetition alone detects this defect. Increasing the inner-solve budget reduces the tower error before it reaches a plateau. This supports a truncation explanation for the tabulated tower result, without attributing every remaining error to the inner solve.

5 Execution cost and convergence

ADMM and projected Gauss-Seidel perform different local operations. Iteration counts consequently cannot substitute for elapsed time. Table D compares executed solves on the same cloud device, recording the budget for each method. PGS uses its iteration-loop timer; synthetic ADMM uses graph replay, whereas lattice ADMM includes residual checking in its forward-loop timer. The timings exclude upload, coloring and spectral setup, so their boundaries are not identical. The synthetic case uses a natural-residual gate; the lattice row compares fixed executed budgets. The lattice residuals come from the corresponding accuracy experiment, not from residual fields in the timing log. Its lower PGS residual means that the ratio of those times is not a matched-accuracy speedup. PGS wins the tabulated synthetic case; ADMM executes the lattice budgets faster. The synthetic ADMM timing excludes the search over penalty candidates, and both timings exclude setup. These rows do not establish a general crossover at matched accuracy.

Spectral penalties can expose a global mass-ratio mode that contact diagonal blocks miss. They do not remove the distinction between sticking, cone binding and near separation. The tested routing rules fail on some packing and nearly open configurations, and selection-set success does not establish generalization. Compliance can improve conditioning while changing the force distribution. We therefore make no universal penalty or compliance-error claim from these experiments.

Table E counts contact-kernel work on pre-exported scenes. A forward environment-step excludes robot dynamics export, collision detection, graph construction and parameter estimation. A sensitivity direction is one prescribed parameter perturbation per environment, not a full Jacobian or a complete simulation step with gradients. The number of directions matters when comparing the robot rows. Rates for the sensitivity phase exclude the forward solve and must not be read as combined forward-and-derivative throughput. The nominal sticking population contains slipping contacts, so the sensitivity workload includes a friction direction as well as a

torque direction. The reported rates are execution measurements; no derivative-error gate accompanies each timed environment.

The cone-binding accuracy row uses a longer budget than its timing row. Throughput at that accuracy point remains unmeasured. The Talos rate measures the contact solve on an L4 using inputs exported beforehand. Dedicated-device and shared-device timings remain distinct, and the latter carry the caveat “GPU delad.” None of these rates is a matched gradient comparison with another simulator.

6 Sensitivity as information

The object experiment estimates box mass, yaw inertia and table friction from a simulated robot pushing a box. Its forward contact solve runs on the GPU, while a separate CPU implementation assembles and solves the implicit branch Jacobian for estimation. The experiment tests the information in contact observations rather than the execution of the GPU sensitivity method. Dynamics and hand kinematics are recomputed at the integrated state. The estimator conditions on measured joint velocities and uses a compliant contact model. It does not feed noisy torque into the free hand velocity, which would change the observation model. Contact candidates are prescribed for this trajectory, and joint positions and velocities are treated as noise-free. These assumptions limit the identification claim independently of the derivative implementation.

Let J_{obs} be the observation Jacobian and σ_{τ} the torque-noise standard deviation. Under independent Gaussian torque errors with common variance, the local information and relative standard-deviation bounds for a locally unbiased estimator with nonsingular information are

$$[F = J_{\text{obs}}^T J_{\text{obs}} / \sigma_{\tau}^2, c_j = \dots]$$

Thus $\sigma_{\tau} c_j$ is a relative standard-deviation bound, not a variance bound. Interpreting these bounds as predictions of observed errors additionally requires the estimator to attain the local covariance bound. The information calculation alone establishes neither unbiasedness nor efficiency. Table G reports absolute parameter errors divided by the corresponding true parameter magnitude, then takes percentiles across environments. It applies the same acceptance threshold to mass and friction; yaw inertia is estimated jointly but is not covered by that acceptance test. Lower noise meets the target, while a longer observation window at the higher noise still fails it.

Stronger excitation changes this information, but the resulting maneuver must remain consistent with the model and the robot’s limits. Table H compares the standard trajectory with the selected excitation under the same population and noise realization. The mass error decreases, whereas the friction target remains unmet. The selected trajectory loses hand contact frequently and exceeds the joint torque limit. Its true-parameter cost and estimator contact residual also rise, so its information score cannot be interpreted as a validated physical experiment. The cost is the sum of squared torque discrepancies across joints, steps and environments with observation noise removed; it is not a mean or a normalized likelihood. The standard trajectory itself exceeds a torque limit; it is not a feasible hardware baseline.

The local sticking corollary explains why friction probing needs a regime change. It does not ensure that the induced slip is observable above sensor noise. Nor does an information score certify contact preservation or actuator feasibility. No physical force or torque sensor validates the identification experiments, and the noise realization has not been replicated across independent noise seeds. A feasible probing policy and material calibration on hardware remain untested.

7 Limits of the integrated simulation

The contact and derivative checks do not establish successful pile dynamics. Correcting a cache that changed contact geometry even at fixed body poses allows the tested tower to satisfy its drift

criterion. Under the corrected contact path, however, piles run from their initial states still fail to settle: PGS leaves unresolved contact residuals, and the tested ADMM configuration diverges. Cleaning the contact set improves conditioning without meeting the residual target. This rules out the identified cache defect as a complete explanation, but does not isolate every remaining geometric or integration error.

Alternative soft-step and barrier-inspired updates also fail to establish rest for the tested piles. Those ports omit or alter parts of the corresponding algorithms, so their failures are not evaluations of the published methods in full. A larger pile was only probed from a saved checkpoint; that experiment measures neither time to rest nor a successful trajectory from the initial state. The contact sensitivities validated here therefore support local calculations within measured regimes, with integrated stability and practical identification requiring separate evidence.

Appendix A. Sliding contact

Table A shows the trajectory maximum normal gap for the same sliding-box configuration under the exact and relaxed contact laws. The relaxed law produces separation despite the matched geometry and iteration budget.

Law	Box side [m]	Mass [kg]	Initial speed [m/s]	μ [-]	Δt [s]	Steps [-]	Sweeps/step [-]	Maximum gap [m]
Coulomb	0.20	1	1	0.3	1/240	200	400	1.552204e-11
Convex relaxation	0.20	1	1	0.3	1/240	200	400	4.352523e-3

Appendix B. Bilateral derivative reference

Table B gives relative entrywise errors against analytical bilateral dynamics in sticking stance configurations, with the torque derivatives scaled from acceleration to velocity. Agreement tests the shared sticking equations; it does not test Coulomb slip.

Robot	Velocity DOF [-]	Contacts [-]	$\delta v^+/\delta \tau$ error [-]	$\delta \lambda/\delta \tau$ error [-]	$\delta v^+/\delta q$ error [-]	$\delta \lambda/\delta q$ error [-]
Unitree A1	18	4	1.22e-13	3.86e-12	2.99e-11	1.62e-10
ANYmal C	18	4	1.27e-12	3.51e-13	7.56e-10	4.75e-10
Talos	38	2	1.75e-14	7.20e-12	1.09e-09	5.23e-10

Appendix C. Slipping derivative reference

Table C reports relative derivative errors for the faulty and corrected kernels at the same reference state. The correction changes the slipping derivatives, including the mixed case whose last contact slips. The compliance multiplier α_η sets $\eta_c = \alpha_\eta \|G_{cc}\|_2$; it is distinct from the natural-map step scale. The reference residual is evaluated with the resulting regularized operator.

Scene	Contacts [-]	Slip [-]	Compliance multiplier [-]	CG iterations [-]	Reference residual [N·s]	$\delta \lambda/\delta b$ error, faulty [-]	$\delta \lambda/\delta b$ error, corrected [-]	$\delta \lambda/\delta \mu$ error, faulty [-]	$\delta \lambda/\delta \mu$ error, corrected [-]
Pushed cube	4	4	0.01	60	9.853e-16	2.179e-01	1.326e-13	9.920e-01	1.427e-13
Mass-ratio column	3	0	0.01	60	2.132e-14	3.401e-13	3.401e-13	0.000e+00	0.000e+00

Scene	Contacts [-]	Slip [-]	Compliance multiplier [-]	CG iterations [-]	Reference r_{nat} [N·s]	$\delta\lambda/\delta b$ error, faulty [-]	$\delta\lambda/\delta b$ error, corrected [-]	$\delta\lambda/\delta\mu$ error, faulty [-]	$\delta\lambda/\delta\mu$ error, corrected [-]
Tower	32	0	0.01	60	1.263e-15	1.751e-09	1.751e-09	0.000e+00	0.000e+00
Sliding chain, one contact	1	1	0.0	60	2.498e-16	3.610e-02	2.220e-16	5.758e-02	1.406e-16
Sliding chain, two contacts	2	2	0.0	60	4.441e-16	3.498e-02	1.243e-15	5.413e-02	2.044e-16
Sliding chain, four contacts	4	4	0.0	60	3.553e-15	3.315e-02	8.120e-15	5.121e-02	1.548e-14
Sliding chain, eight contacts	8	8	0.0	60	7.105e-15	3.020e-02	6.532e-14	5.018e-02	2.401e-12
Mixed chain, stick first	4	3	0.0	60	1.332e-15	3.166e-02	5.329e-15	4.576e-02	1.747e-14
Mixed chain, stick last	4	3	0.0	60	1.776e-15	1.104e-14	1.104e-14	2.044e-14	2.044e-14

Appendix D. Executed solve costs

Table D compares PGS and ADMM on an L4. PGS takes less time for the synthetic case, while ADMM executes the stated lattice budget faster. The lattice residuals are from a separate accuracy experiment, so that row does not pair elapsed time with an observed stopping residual.

Scene	Contacts [-]	Stopping specification	PGS sweeps [-]	ADMM iterations [-]	PGS [ms]	ADMM [ms]	PGS r_{nat} [N·s]	ADMM r_{nat} [N·s]
Synthetic	256	$r_{\text{nat}} < 1e-8$ N·s	200	275	72.44	188.66	2.83e-9	<1e-8
Lattice	40000	Fixed budgets	1000	25	445.61	84.09	7.3e-17	2.4e-15

Appendix E. Batch budgets and rates

Table E separates contact-kernel throughput from the fraction below the natural-residual tolerance, with a dash denoting an unreported quantity at that budget. Increasing the cone-binding budget still leaves part of the population above tolerance. Rates are counts of executed directions, not counts of independently validated derivatives.

Population	Device	Environments [-]	ADMM budget [-]	r_{nat} gate [N·s]	Fraction below gate [-]	Forward [env-steps/s]	Directions /env [-]	Sensitivity [env-directions/s]
A1, nominal stick	RTX 5070; GPU delad	4096	200	1e-8	0.9985	18 320	2	128 350
A1, cone binding	RTX 5070; GPU delad	4096	200	1e-8	—	9 559	2	68 488
A1, cone binding	RTX 5070; GPU delad	4096	2000	1e-8	0.7766	—	—	—

Population	Device	Environments [-]	ADMM budget [-]	r_{nat} gate [N·s]	Fraction below gate [-]	Forward [env-steps/s]	Directions /env [-]	Sensitivity [env-directions/s]
ANYmal C, nominal stick	L4	4096	200	1e-8	0.9971	16 703.5	2	79 506.6
Talos, stick	L4	4096	200	1e-8	1.0000	4 919.6	1	72 009.8

Appendix F. Repeated batch bytes

Table F gives full SHA-256 digests of the concatenation $(\lambda, v^+, \delta\lambda/\delta\tau, \delta v^+/\delta\tau, \delta\lambda/\delta\mu)$, in that order. Each array is converted to contiguous float64 bytes before hashing. Equality across the listed devices certifies these recorded outputs, including any derivative error. The identity runs use their own forward budgets and are not the timed runs in Table E.

Batch case	Environments [-]	ADMM budget [-]	L4 = A10G = H100: SHA-256
A1, nominal stick	4096	200	4d5d4747983a3de20aed4d49b12f453ced005275bce8edd94e956d4aa9c3c7d3
A1, cone binding	4096	500	24a370fedefcc7a7b2a3dddb7ebe1bc2432a0259bc2c2265304cb4db1bfd9bf6

Appendix G. Identification under observation noise

Table G reports relative parameter errors over the simulated population and applies the acceptance test that mass and friction each meet the listed percentile threshold. The longer observation window reduces errors but still fails the target at the higher noise level.

Trajectory	Environments [-]	Window [steps]	σ_τ [N·m]	Mass error p50 [-]	Mass error p95 [-]	Friction error p50 [-]	Friction error p95 [-]	p95 threshold [-]	Both pass
Standard push	512	50	0.1	4.0476e-03	1.5272e-02	4.3778e-03	1.6340e-02	0.05	pass
Standard push	512	50	1.0	3.8590e-02	1.5173e-01	4.1072e-02	1.6086e-01	0.05	fail
Standard push	512	100	1.0	2.2260e-02	8.9702e-02	1.9507e-02	8.3003e-02	0.05	fail

Appendix H. Excitation and model consistency

Table H compares the standard push and selected excitation on the same population, distinguishing the generator’s contact residual from the estimator’s residual at the true parameters. The residual columns report maxima over the evaluated contacts, environments and time steps. Lower mass error accompanies increased model discrepancy, lost contact and a torque-limit violation; the friction target remains unmet.

Trajectory	Environments [-]	Window [steps]	σ_τ [N·m]	Mass error p95 [-]	Friction error p95 [-]	True-parameter squared torque cost [(N·m) ²]	Estimator r_{nat} [N·s]	Generator r_{nat} [N·s]	No hand contact [fraction]	Peak joint torque / limit [-]
Standard push	512	50	1.0	1.517289e-01	1.608568e-01	2.347430e-06	1.282599e-06	4.199e-12	0.0000	1.154

Trajectory	Environments [-]	Window [steps]	σ_τ [N·m]	Mass error p95 [-]	Friction error p95 [-]	True-parameter squared torque cost [(N·m) ²]	Estimator r_nat [N·s]	Generator r_nat [N·s]	No hand contact [fraction]	Peak joint torque / limit [-]
Selected excitation	512	50	1.0	3.818469e-02	8.413782e-02	1.503296e+01	1.481804e-02	2.161e-08	0.7213	5.284

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